Jan Arogya Pusthak (JAP)

A Unified Digital Health Ledger for Indian Citizens

I. Terminology & Country Context

Pre-existing conditions (PECs): In the Indian healthcare context, this refers to the chronic illnesses like hypertension and diabetes, which make up the vast majority of outpatient visits in primary health centers and small private clinics.

Ayushman Bharat Health Account (ABHA): A unique 14-digit digital identity card assigned to Indian citizens. (Data Baseline: As of mid-2025, over 79.7 crore ABHA accounts have been created, with more than 65 crore health records linked via the National Health Authority dashboard).[1]

Jan Arogya Pusthak (JAP): A lightweight user interface built directly over the Ayushman Bharat Digital Mission (ABDM) framework. Its primary innovation is removing the barrier of smartphone app installations or complex application management by enabling access via a standard Aadhaar number and a secure PIN.

II. Executive Summary

Vision: To provide every Indian citizen with a lifelong, portable, paperless health passbook accessible instantly via the cloud or standard SMS text messaging.

Mission: To build a lightweight, UPI-like digital health utility on top of the existing Ayushman Bharat Digital Mission (ABDM) ecosystem to maximize accessibility and lower usage friction.

Strategic Thesis:

- Underfunded Public Health: India's public health expenditure stands at 1.9% of GDP (FY 2025-26), falling short of the 2.5% target outlined in the National Health Policy (2017)[2]. The Union government's share is just 0.28% of GDP.
- Financial & Resource Waste: The healthcare infrastructure loses billions of rupees annually due to duplicate, redundant medical testing caused by lost or damaged paper records. Out-of-pocket (OOP) diagnostic costs account for up to 70% of total hospitalization bills.[3]
- The Adoption Gap: While 79.7 crore ABHA IDs exist, only 65 crore are actively linked to medical records. Existing applications (like the ABHA App or Aarogya Setu) demand stable smartphones, storage space, and confusing logins, making them difficult to scale across rural populations.[4]

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[1] PIB - Update on ABDM, 12th August, 2025
[2] [IBEF Report (Feb 2025)]
[3] PIB - Change in Medical Expenditure Patterns, 11th February, 2020
[4] ABDM in rural areas LS-Unstarred Question No-2110, To be answered on 01.08.2025

The Solution:
Jan Arogya Pusthak (JAP) simplifies this dynamic completely. Patients carry no physical records. Doctors retrieve patient medical history within seconds using only the patient's Aadhaar number and PIN. Consulting summaries are delivered instantly via SMS. The platform utilizes zero-cost public architecture (DigiLocker and MeghRaj cloud) and is incentivized through the government's Digital Health Incentive Scheme (DHIS).

III. Problem Deep Dive

- Operational Vulnerability: Rural healthcare suffers from extreme fragmentation. The doctor-to-patient ratio is approximately 1:25,000 in several regions, cutting consultation windows short.[5] Finding a patient's medical history files may take around 5 to 10 minutes. In practice in most cases it does not exist, forcing physicians to make diagnoses based on incomplete data and assumptions.

- Information Asymmetry: Administering medical care without historical context poses critical clinical risks, particularly for common ailments like diabetes and hypertension, which currently go undiagnosed in over half of affected individuals.[6]

- Economic Burden: Missing records force repetitive diagnostic screenings. These redundant diagnostics consume over 70% of a family's out-of-pocket medical expenditure, driving low-income households further into poverty.[7]

IV. User Jobs to Be Done

User Role	Core Need (Job to be Done)
Patient	"When I visit a new doctor, I want them to know my past medical conditions automatically without requiring me to memorize them or carry stacks of paper files."
Doctor	"When a patient sits across from me, I need immediate visibility into their allergies, chronic conditions, and recent prescriptions so I can make quick, data-backed decisions instead of relying on guesswork or family recollections."
Nurse/Lab Tech	"I want to log vital signs and upload laboratory results efficiently without maintaining redundant physical registers, ensuring information is preserved perfectly for the primary doctor."

[5] Bridging the urban-rural divide in India's diagnostic market, Medical Buyer, Dec 2025
[6] Comprehensive approach to diabetic care, The ET, 6th December 2024
[7] PIB - Change in Medical Expenditure Patterns, 11th February, 2020

V. Target Market & User Personas

1. **Primary Market**

   Facility type: Government Primary Health Centers (PHCs), Community Health Centers (CHCs), small neighborhood clinics, and district hospitals struggling to maintain structured health records.

   Demographics: Low-income families, daily wage laborers, senior citizens, and individuals unfamiliar with complex medical or digital terminology.

2. **User Personas**

Persona	Demographics	Goals	Pain Points	Success Metric
Patient	Farmers, daily wage earners, senior citizens, migrant workers.	Retain permanent medical logs; Get efficient treatment without heavy over-explanation.	Forgetting past health stats (e.g., BP trends, medication names); Losing physical papers.	Receives prescriptions and lab summaries directly via SMS; Accessible via Aadhaar + PIN.
Doctor	Government PHC/CHC physicians, private practitioners, etc	Execute rapid, precise diagnoses referring to high-quality historical medical data.	Total lack of patient charts; tedious manual entries; doctors unaware of patients' medical context.	Accesses full patient history in ≤ 30 seconds following a simple secure PIN entry.
Nurse, Lab Tech	Clinic nurses, lab assistants, triage staff.	Log patient vitals without error; store and pull up patient data efficiently.	Balancing multiple physical paper registers; time-consuming document retrieval.	Data is permanently saved to the cloud, eliminating manual registers and paper loss.

VI. Quantitative Validation and Baseline Metrics

Metrics	Estimated Baseline	Target Improvement
% of OPD patients with valid historical records available at consultation	<15%	>70%
Average time required for a doctor to retrieve a patient's history	5-10 min (if files are there)	<30 sec
% of a family's out-of-pocket medical expenditure on diagnostic tests	>70%	<30%

Metrics	Estimated Baseline	Target Improvement
Digital health record adoption (any form) in rural India	<1%	>30%

VII. Competitive Landscape

Solutions	Type	Strength	Weakness	Why JAP?
ABHA App - Ayushman Bharat Health Account	Official Govt App	Comprehensive, highly secure, fully integrated feature set.	Requires a smartphone, steady internet connectivity, storage space, and complex logins.	Zero Friction: Operates with an ATM-like PIN and delivers summaries straight via basic SMS.
Paper records	Traditional	No technological learning curve; highly familiar.	Easily lost, torn, damaged, or left behind; completely unportable.	Total Portability: Zero physical documents to carry or preserve.
Clinic-Specific EMRs	Paid Proprietary Software	High feature density; built for enterprise hospital systems.	Expensive licensing fees; lacks cross-facility interoperability.	Unified Ecosystem: Free to use, open-source architecture, and works across all clinics.

Note: JAP serves as an administrative delivery layer built over the ABDM framework; it is a feature enabler, not a competitor to the ABHA ID ecosystem.

VIII. Public Policy & Compliance Framework

1. **Digital Personal Data Protection (DPDP) Act, 2023 Compliance**
   JAP acts strictly as a Data Fiduciary under the DPDP framework:
   - ***Notice & Consent:*** The patient's secure PIN input functions as a static key consent, granting the doctor temporary verification to access records.
   - ***Data Retention:*** Follows sensitive personal data guidelines; clinical records are locked against unauthorized erasure to maintain diagnostic truth.

2. **NABH Digital Health Standards**
   JAP complies with the National Accreditation Board for Hospitals & Healthcare Providers (NABH) standards for Health Information Systems (HIS), ensuring

interoperability, data safety, and immediate eligibility for government funding via the Digital Health Incentive Scheme (DHIS).

3. **Integration with Ayushman Bharat Digital Mission (ABDM)**
   The ABHA ID remains the master key. JAP does not generate a new identity system; it acts as an agile retrieval tool aligned directly with India's national federated digital health layout.

IX. Product Principles

The following product principles guide every design and prioritization decision:

a. ***Zero App Downloads*:** No application installation or smartphone dependency for the patient.

b. ***Rapid Checkouts*:** Doctors can complete and finalize a full digital consultation entry in ≤ 90 seconds.

c. ***Instant History*:** Patient medical data must load in ≤ 30 seconds after Aadhaar verification and PIN input.

d. ***Offline-First Stability*:** Core tools (Aadhaar lookup, PIN confirmation, local data caching) run efficiently on low-bandwidth networks.

e. ***Data Sovereignty*:** Medical history can never be downloaded, altered, or deleted by external third parties without patient authorization.

f. ***Immutable Logs*:** Every addition or modification creates an unalterable audit trail ensuring Good Clinical Practice (GCP) transparency.

g. ***Tamper-Proof Files*:** Documents feature built-in cryptographic signatures and exact timestamps.

h. ***Universal Accessibility*:** Fully supports low digital literacy cohorts via feature phones and basic SMS.

X. Minimum Viable Product

1. **Doctor & Nurse Web Portal (Desktop/Tablet)**
   - Registration and Verified Onboarding: Registration of Account along with National Medical Council/State council verification valid for practising.
   - Patient Lookup: Rapid patient charting via physical Aadhaar card scanning or manual number input followed by patient PIN entry.
   - Patient Health Timeline View: A clean, chronological, chat-style interface presenting historical diagnoses, medications, and vitals.
   - Smart Rapid Data Entry System: Dynamic dropdowns and intuitive fields to quickly log vitals, allergies, blood groups, and diagnostic notes.
   - SMS Transmission: A one-click interactive button that translates the digital prescription into an immediate text summary sent to the patient's phone.
   - Digital Sign-off: Consultations and tests automatically apply the provider’s digital signature and timestamp.

2. **Patient Facing (Website)**
   - Registration and PIN: Registration of Account along with creation of PIN.
   - Read-Only Ledger: Secure login using Aadhaar and PIN to view or download historical health logs without any ability to modify original clinical records.

XI. Go-to-Market (GTM) Strategy

1. ***Institutional Alignment***: Partner with District Medical Officers (DMOs) to replace paper registers with digital interfaces at public health facilities.
2. ***Private Practitioners Onboarding***: Introduce the tool to standalone neighborhood doctors and local private practices.
3. ***Grassroots Enrollment***: Coordinate with local NGOs to deploy account setup booths at free medical camps, community centers, and mini-clinics.

XII. Key Performance Indicators (KPIs)

KPI Category	Metric	Target
Adoption	Total unique patients with ≥ 1 JAP records created via SMS or web	20% of the target regional or district population
Engagement	Active unique patients accessing records within a rolling 30-day window	> 30% of total registered user database.
Operational	Percentage of outpatient consultations leveraging JAP history	> 60% of total OPD consultations at a facility
Value	Percentage of doctors reporting a time savings of ≥ 2 minutes per patient	> 75% of participating doctors.
Value	% of a family's out-of-pocket medical expenditure on diagnostic tests	< 30% of the out-of-pocket medical expenditure.

XIII. Scaling & Future Roadmap

1. Value-Add Ecosystem: Introduce premium provider tiers incorporating integrated telemedicine workflows and AI-driven medical history summaries.
2. Research & Analytics: Supply de-identified, anonymized health datasets to government bodies, medical universities, and pharmaceutical companies to support epidemiological studies and public health drug research.